Plotting & Visualisation

Arguments that can be passed into plot():

(R Studio default.)

- color / col
- alpha = transparency or transp ← using epiEstim package
- xlab/ylab = change x-axis
- n_breaks = # of breaks on x-axis
- What = what data to plot (e.g: all, incid, R, SI)
- x = output function
- Xlim/Xlim = x and y limits.
- interval = time intervals for plotting incidence - default 1 day

Plotting Principals:

- Colour-blind friendly? Prints well in b+w? Simple? axes labelled? font & labels readable?

ggplot comes with tidyverse

Benefits:

+ Assigning plots to objects = save to .RDS not image so can edit later
+ wide varity of options.
+ User friendly
+ Outputs look more proffesional than using 'base' R functions

GG plot object layers:

%>% ggplot(...) + geom_type(.) + Scale_what_how(.) + Stat_method(.)

- geom_type: what type of plot
- Scale_what_how: manipulate scale
- Stat_method: transform data
- ggplot(...): either must contain aesthetics (aes) - which vars + how

Using aesthetics:

ggplot (aes (x = x_val, y = y_value, color = what))

eg: sex
male or female
colour different

or geom_line → gives you line graph

Faceting = dividing plot into subplots

theme_minimal (base_size = _, base_family = _, etc)

→ change text

→ font size

→ font family

lty = line type
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